Mid-Semester Research Report

Evaluating the Stability, Fidelity and Biological Plausibility of Feature Attributions in Deep Learning-Based Virus Classification

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Abstract

Deep learning models for viral genomic classification now achieve high accuracy, yet they rarely reveal which genomic features drive a prediction. High accuracy can hide reliance on sequencing artefacts, composition bias, or database effects, leaving biological grounding unproven. This project evaluates the *stability*, *fidelity*, and *biological plausibility* of feature attributions generated by three deep learning architectures (CNN, BiLSTM, and a custom Transformer) trained on a leakage-free dataset of six RNA and DNA viruses. Four attribution methods (Occlusion, Permutation, Integrated Gradients, and KernelSHAP) are compared, and a reusable, model-agnostic evaluation pipeline is delivered through an interactive FastAPI dashboard with 3D protein mapping. Current findings show that even the highest-accuracy model (Transformer, 96% F1) often produces unstable, low-fidelity, and biologically implausible explanations on held-out lineages.

1 Problem and Research Objective

Deep learning tools for virus classification and discovery now perform well on genomic benchmarks, yet they rarely show which genomic features drive a prediction. High accuracy can hide reliance on artefacts such as sequence length, nucleotide composition, database bias, or sequencing platform effects. In clinical virology and public health surveillance, an explanation that is not biologically grounded is of limited use: a model that confidently labels a sequence as Dengue while attending to intergenic noise is not trustworthy, even if its overall F1 score is high.

The central problem is therefore not classification accuracy but **explanation validity**: we lack systematic methods to determine whether the genomic regions a model highlights are (a) stable across architectures, (b) faithful to the model's actual decision, and (c) biologically plausible.

General Objective

To evaluate the stability, fidelity, and biological plausibility of feature attributions generated by deep learning-based viral genomic classification tools.

Specific Objectives

1. Train multiple deep learning architectures (CNN, BiLSTM, and a Transformer) on a diverse, leakage-free dataset of RNA and DNA viruses.
2. Implement four post-hoc attribution methods (Occlusion, Permutation, Integrated Gradients, and SHAP) and compare their outputs.
3. Quantify **stability** as cross-architecture agreement in attribution rankings.
4. Quantify **fidelity** as the drop in predicted probability when top-attributed regions are masked.
5. Quantify **biological plausibility** as the overlap between top-attributed regions and annotated genes.
6. Deliver a reusable, model-agnostic evaluation pipeline and an interactive dashboard that makes the audit reproducible and inspectable.

2 Literature Review and Research Gap

2.1 What Exists

Bartoszewicz et al. [**?**] visualised convolutional filters for human-virus detection from NGS reads; EdeepVPP [**?**] extracts learned filters from a CNN to highlight informative viral patterns. A 2026 *Scientific Reports* study [**?**] applied SHAP to a SARS-CoV-2 variant CNN and compared it with a chi-square GWAS baseline; SHAP flagged lineage-associated Spike sites as well as ORF8, ORF9, and intergenic positions the baseline missed. Methodologically, the field draws on fidelity benchmarks such as Hooker et al. [**?**] and axiomatic attribution such as Integrated Gradients [**?**].

2.2 The Gap

No standard routine compares attributions **across deep learning architectures** and tests them against **annotated biology across virus families**. Existing studies typically apply a single attribution method to a single model for a single virus. They do not ask:

- Do different architectures agree on which regions matter?
- If the highlighted regions are removed, does the prediction actually break?
- Do the highlighted regions overlap with known genes more often than chance?
- Can these checks be run reproducibly on new sequences?

2.3 Proposed Contribution

A reusable, model-agnostic pipeline delivering:

1. A **consensus feature ranking** across architectures, with a stability score.
2. **Fidelity tests** via deletion and masking, with controls.
3. A **per-family biological plausibility score** comparing attribution peaks to annotated genes.

4. An **interactive dashboard** that runs the entire audit live on an uploaded sequence, including 3D mapping of attributed regions onto protein structures.

3 Methodology and System Architecture

This section provides a reproducible technical breakdown of the Viral XAI pipeline. The system takes raw viral genomic sequences, classifies them using three deep learning architectures, extracts feature attributions using four XAI methods, and evaluates those attributions across three scientific pillars (Stability, Fidelity, Plausibility).

3.1 Phase 1: Data Acquisition, Curation, and Splitting

Source Databases

- **Taxonomic Ground Truth:** The ICTV Virus Metadata Resource (VMR) [**?**] was used to establish the authoritative lineage of each virus.
- **Sequence Data:** Complete genomes were fetched programmatically from NCBI GenBank via Entrez Direct (`efetch`) and the NCBI Datasets CLI.

The Lineage-Level Holdout Strategy

Random and temporal splits resulted in data leakage because sequences within the same lineage remain highly similar over time (a 2015 and 2023 Dengue genome can be 99% identical). To guarantee zero leakage, we implemented strict **Lineage-Level Holdouts**. The final scope was narrowed to six viruses (three RNA, three DNA), totaling 6,043 complete genomes.

Virus	Genome	Training Lineages	Test Holdout
Dengue	RNA (+ssRNA)	DENV-1, DENV-2, DENV-3	**DENV-4**
Chikungunya	RNA (+ssRNA)	ECSA genotype	**Asian genotype**
Norovirus	RNA (+ssRNA)	GII, GV, GIX	**Genogroup GI**
Measles	RNA (-ssRNA)	D8, D4	**Genotype B3**
Mpox	DNA (dsDNA)	Clade Ia, IIa, Ib	**Clade IIb**
Adenovirus C	DNA (dsDNA)	Types 2, 5	**Type 1**

Table 1: Lineage-level holdout strategy for six viruses.

Result: 5,073 training sequences and 970 testing sequences. Sequences were split into overlapping 3,000 bp windows (stride 1,500 bp).

3.2 Phase 2: Deep Learning Architectures

Three models were trained from scratch on CPU-only hardware to evaluate cross-architecture stability.

Model 1: Convolutional Neural Network (CNN)

- **Purpose:** Captures local genomic motifs.
- **Architecture:** Two 1D convolutional layers (64 and 128 filters, kernel size 8), batch normalization, max pooling, and a dense output layer.
- **Performance:** 75% test F1-score.

Model 2: Bidirectional LSTM (BiLSTM)

- **Purpose:** Captures sequential dependencies by reading the sequence in both directions.
- **Architecture:** 64-dimensional embedding layer, bidirectional LSTM (64 hidden units), mean pooling, and a dense output layer.
- **Performance:** 86% test F1-score.

Model 3: Custom Genomic Transformer

- **Purpose:** Captures global genomic context. Pre-trained models like DNABERT-2 were abandoned due to Triton/CUDA dependency conflicts on CPU.
- **Architecture:** 128-dimensional embedding layer, a 2-layer Transformer encoder (4 attention heads, 256 feed-forward dimension), mean pooling, and a dense output layer.
- **Performance: 96% test F1-score** (best on this dataset).

3.3 Phase 3: XAI Attribution Methods

Four distinct attribution algorithms were implemented to calculate how much each 100 bp genomic region contributed to the final prediction.

1. **Occlusion:** A 100 bp window is masked with "N" characters. The drop in predicted class probability is recorded as the importance score.
2. **Permutation:** A 100 bp window is randomly shuffled, preserving nucleotide composition but destroying motif structure.
3. **Integrated Gradients (IG):** Uses gradients of the model, interpolating from a baseline of 25% A/C/G/T to the input.
4. **KernelSHAP:** Uses Shapley values from cooperative game theory, sampling coalitions of masked regions to estimate marginal contributions.

3.4 Phase 4: The Three-Pillar Evaluation

This is the core scientific contribution of the project.

Pillar 1: Stability (Cross-Model Agreement)

- **Method:** Extract the top 5 highest-attributed 100 bp regions from each model.
- **Metric:** Spearman rank correlation (ρ). A score near 1.0 means all models look at the same coordinates; near 0.0 means they look at entirely different regions.

Pillar 2: Fidelity (Deletion Test)

- **Method:** Mask the consensus top 5 regions and re-run the prediction.
- **Metric:** Percentage drop in confidence. A large drop indicates the model was faithful to the explanation; a small drop indicates redundancy.

Pillar 3: Biological Plausibility (Gene Mapping)

- **Method:** Fetch ground-truth CDS annotations for the specific virus from NCBI GenBank.
- **Metric:** Observed vs. Expected (O/E) ratio. An O/E > 1.0 indicates the AI is biologically grounded.

3.5 Phase 5: Interactive Dashboard Deployment

The pipeline is packaged into a web application.

- **Backend:** Python FastAPI, with asynchronous background tasks to prevent UI freezing during XAI computation.
- **Frontend:** HTML/CSS/Jinja2, with Plotly.js for 2D charts (stability lines, fidelity waterfall, plausibility gene track, method agreement heatmap).
- **3D Viewer:** `3Dmol.js` fetches live PDB structures and maps nucleotide coordinates to amino acid residues using $(N_{pos} - CDS_{start})/3$.

4 Progress So Far

4.1 Completed Work

- **Literature:** PRISMA 2020 systematic review completed (18 primary studies).
- **Data:** ICTV release fixed; six viruses with lineage-level holdouts; 5,073 training / 970 test sequences.
- **Models:** CNN (0.75 F1), BiLSTM (0.86 F1), Transformer (0.96 F1) trained on CPU.
- **Attribution:** Four methods implemented (Occlusion, Permutation, IG, KernelSHAP).
- **Evaluation:** Stability, fidelity, and plausibility metrics computed.
- **Dashboard:** FastAPI app with Nextclade-inspired UI, 2D charts, 1D gene track, and 3D protein viewer with gold consensus halos.

4.2 Current Findings

Even the high-accuracy Transformer often produces unstable (low Spearman ρ), low-fidelity (small confidence drop after masking), and biologically implausible (O/E < 1.0) explanations on held-out lineages.

4.3 Limitations Acknowledged

- High/low thresholds are **provisional**; negative controls are needed to calibrate them.
- Occlusion is the default attribution for primary visualisations; other methods appear only in the heatmap.
- The study covers six viruses and may not generalise to all families.
- Bootstrap stability and retraining-without-top-features controls are planned.

5 Remaining Work

- **W1–W2:** Run negative controls (shuffled genomes, randomised labels) to calibrate thresholds.
- **W3:** Compute bootstrap stability.
- **W4:** Finalise dashboard (method toggles, accessibility, user testing).
- **W5:** Complete written report (Chapters 1–6).
- **W6:** Produce poster and finalise prototype demonstration.

5.1 Expected Prototype

A small interactive application where a user uploads a sequence and sees the predicted taxon, top attributed regions mapped to genes, agreement across methods, the three pillar scores, and a 3D protein view with consensus residues highlighted.

6 End-to-End System Flowchart

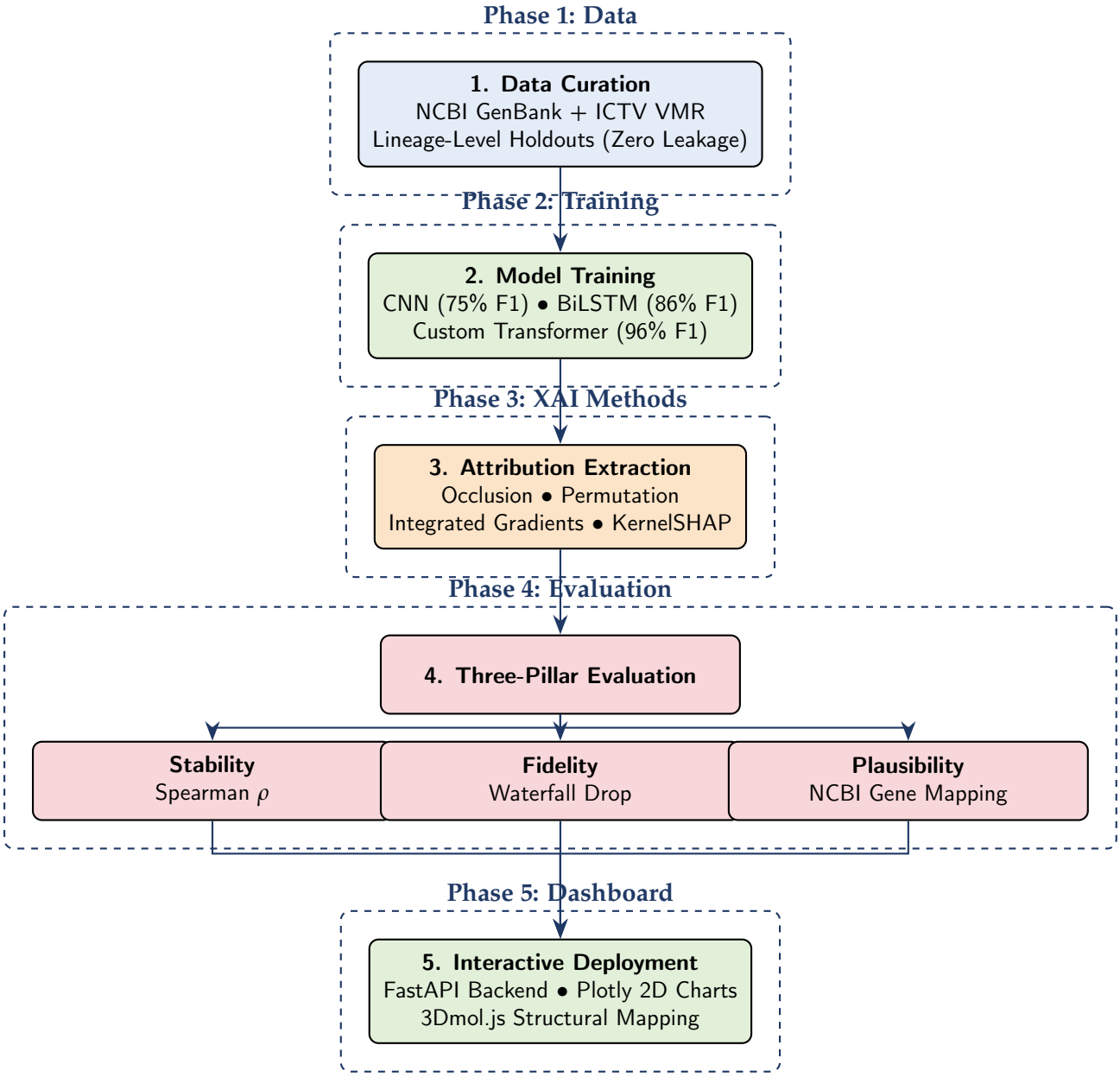


Figure 1: End-to-end technical architecture of the Viral XAI pipeline.